

18 March, 2-3 April 2013

Tutorial Test 4

**2-D Motion /2-D
*beweging***

Monday: Question 10, and Problem 29

Tuesday: Question 8 a,b and Problem 97

Wednesday: Question 8 c,d and Problem 104

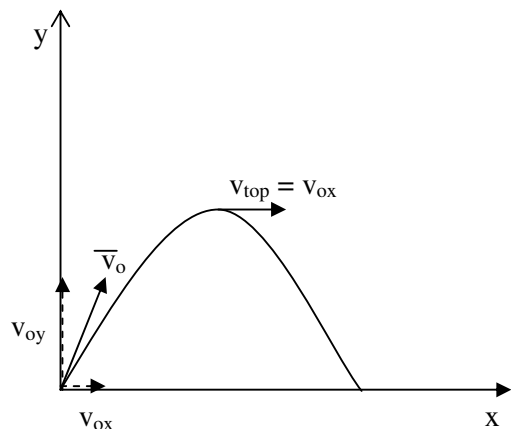
Monday:**Question 10**

(a) c, b, a

✓✓

(b) a, b, c

✓✓

Problem 29**A Projectile's launch speed is five times its speed at maximum height. Find the launch angle θ_0 .**

The coordinate origin is at its initial position (where it is launched).

At maximum height, we observe $v_y = 0$ ✓ and denote $v_x = v_{top}$ (which is also equal to v_{0x}). ✓

In this notation, we have $v_0 = 5v$. ✓

Next, we observe $v_0 \cos \theta_0 = v_{0x} = v$ ✓, so that we arrive at an equation (where $v \neq 0$ cancels) which can be solved for θ_0 :

$$(5v) \cos \theta_0 = v$$

$$\Rightarrow \theta_0 = \cos^{-1} \left(\frac{1}{5} \right) = 78.5^\circ. \quad \checkmark \checkmark$$

If the student has done these solutions using another method – which MAY be correct and gets the same answers, please distribute the marks accordingly.

Thank you.

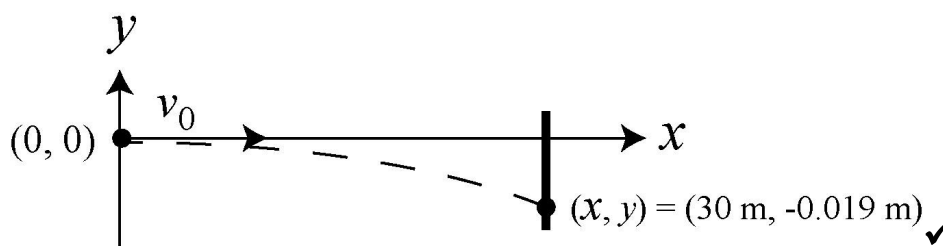
Tuesday:**Question 8 a,b**

- (a) 3, 2, 1 passes window from bottom to top – vertical speed will play a role ✓✓
 (b) 1, 2, 3 ✓✓

Problem 97

A rifle is aimed horizontally at a target 30 m away. The bullet hits the target 1.9 cm below the aiming point. What are the bullet's time of flight and (b) its speed as it emerges from the rifle?

The trajectory of the bullet is shown in the figure below (not to scale). Note that the origin is chosen to be at the firing point. With this convention, the y coordinate of the bullet is given by $y = -\frac{1}{2}gt^2$. Knowing the coordinates (x, y) at the target allows us to calculate the total flight time and speed of the bullet.



- (a) If t is the time of flight and $y = -0.019 \text{ m}$ indicates where the bullet hits the target, then from the equation: $y = v_{0y}t - \frac{1}{2}gt^2 = -\frac{1}{2}gt^2$ ✓

$$t = \sqrt{\frac{-2y}{g}} = \sqrt{\frac{-2(-0.019 \text{ m})}{9.8 \text{ m/s}^2}} = 6.2 \times 10^{-2} \text{ s.} \quad \checkmark$$

- (b) The muzzle velocity is the initial (horizontal) velocity of the bullet. Since $x = 30 \text{ m}$ is the horizontal position of the target, we have $x = v_{0x}t + \frac{1}{2}a_x t^2 = v_{0x}t$. ✓ Thus,

$$v_0 = \frac{x}{t} = \frac{30 \text{ m}}{6.3 \times 10^{-2} \text{ s}} = 4.8 \times 10^2 \text{ m/s.} \quad \checkmark \checkmark$$

Alternatively, we may use the trajectory (path) equation $y = \tan \theta_0 - \frac{gx^2}{2(v_0 \cos \theta_0)^2}$ ✓ to solve for the initial velocity.

With $\theta_0 = 0$ and $y_0 = 0$, the equation simplifies to $y = -\frac{gx^2}{2v_0^2}$, leading to

$$v_0 = \sqrt{-\frac{gx^2}{2y}} = \sqrt{-\frac{(9.8 \text{ m/s}^2)(30 \text{ m})^2}{2(-0.019 \text{ m})}} = 4.8 \times 10^2 \text{ m/s} \quad \checkmark \checkmark$$

which is precisely what we calculated in part (b).

If the student has done these solutions using another method – which MAY be correct and gets the same answers, please distribute the marks accordingly. Thank you.

Wednesday:**Question 8 c,d**

- (a) all the same (passes window from left to right – horizontal speed is the same for all cases.) ✓✓
 (b) 6, 5, 4 ✓✓

Problem 104 A ball is thrown horizontally from a height of 30 m and hits the ground with a speed that is three times its initial speed. What is the initial speed?

The initial velocity has magnitude v_0 and because it is horizontal, it is equal to $v_{0x} = v_x$ ✓ the horizontal component of velocity at impact. Thus, the speed at impact is

$$\sqrt{v_0^2 + v_y^2} = 3v_0 \quad \dots\dots\dots 1 \checkmark$$

Since $v_y^2 = v_{0y}^2 + 2a\Delta y$ and $v_{0y} = 0$ m/s, then $v_y = \sqrt{2gh}$, ✓ where $\Delta y = h = 30$ m. Squaring both sides of equation 1, and substituting for v_y from above:

$$v_0^2 + 2gh = (3v_0)^2 \quad \dots\dots\dots 2 \checkmark$$

which leads to $gh = 4v_0^2$ and solving for v_0 then gives:

$$v_0 = \sqrt{(9.8 \text{ m/s}^2)(30 \text{ m}) / 2} = 8.6 \text{ m/s.} \checkmark \checkmark$$

If the student has done these solutions using another method – which MAY be correct and gets the same answers, please distribute the marks accordingly .

Thank you.